

CAPSTONE PROJECT

Social Media Sentiment & Engagement Analysis

Executive overview of customer sentiment, activity and engagement patterns

732	33	3
Total Posts	Countries	Platforms

Power BI | Power Query | DAX

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Overview • Analysis • Insight

Project Overview

This capstone project analyzes social media activity to understand customer sentiment, engagement, platform performance, geographic activity and time-based patterns. It was completed as the capstone of my Business Analytics Internship at Codveda Technologies, using Power BI, Power Query and DAX.

Business Problem

The company receives a large volume of social media activity across platforms and markets but lacks a clear view of customer sentiment and engagement. Without it, the company cannot easily identify high-performing platforms, understand customer reactions, monitor sentiment risks, or tell when and where engagement is strongest.

Objectives

- Analyze customer sentiment and its link to likes and retweets
- Compare platform performance
- Examine geographic activity
- Identify monthly and hourly activity patterns
- Turn findings into practical recommendations to improve engagement

Dataset and Tools

The dataset contains **732** social media records covering **33 countries** and **3 platforms** (Instagram, Twitter, Facebook), with posts dated from 2010 and a sharp rise in 2023. Fields include Sentiment, Platform, Country, Likes, Retweets, Timestamp, Hashtag, Year, Month, Day and Hour.

- **Power Query:** data cleaning and transformation
- **DAX:** analytical measures such as likes and retweets per post
- **Power BI:** dashboard and visualization

Analytical Approach

Business Problem → Data Cleaning → Transformation → Analysis → Dashboard → Deeper Analysis → Insights → Recommendations

The analysis compared **volume** (total posts and likes) with **engagement intensity** (average likes per post). Totals show where activity happens; averages show where content performs.

Dashboard Overview

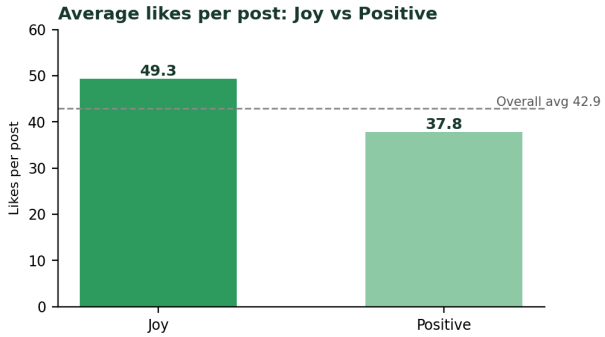
The Power BI dashboard has four pages: **Cover** (project title and headline numbers), **Overview** (KPIs, sentiment distribution, posts by platform, activity over time with year drill-down), **Analysis** (country, country and sentiment, activity by hour) and **Insight** (findings, actions and success measures).

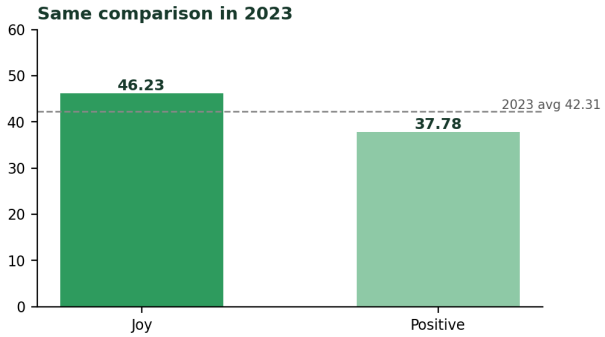
Headline numbers: 732 total posts, 42.90 average likes per post, 21.51 average retweets per post, Joy as the top engaging sentiment, the USA as the most active country and 2 PM (hour 14) as the peak activity hour.

Key Results & Business Insights

1. Sentiment & Engagement

Positive was the most frequent sentiment (45 posts), but Joy earned more per post at about 49.3 likes, compared with 37.8 for Positive. The gap held in 2023, the peak year, at 46.23 vs 37.78 likes.

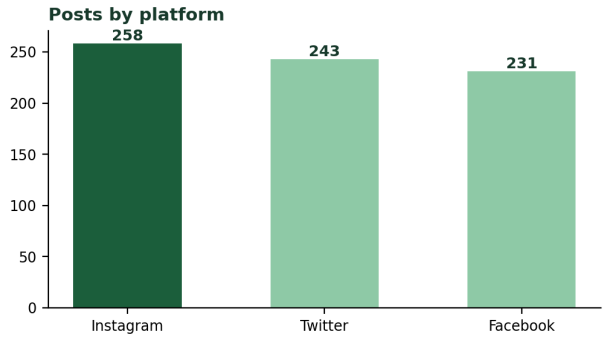


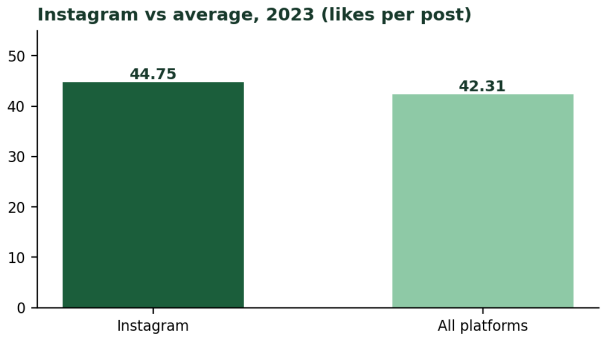


Result: The most common sentiment is not the most engaging, and the finding is not a timing effect.

2. Platform Performance

Instagram led with 258 posts and 11,633 likes, and had the highest average of about 45.1 likes per post. It also stayed above average in 2023 (44.75 vs 42.31). Twitter (243 posts) and Facebook (231 posts) followed on volume.

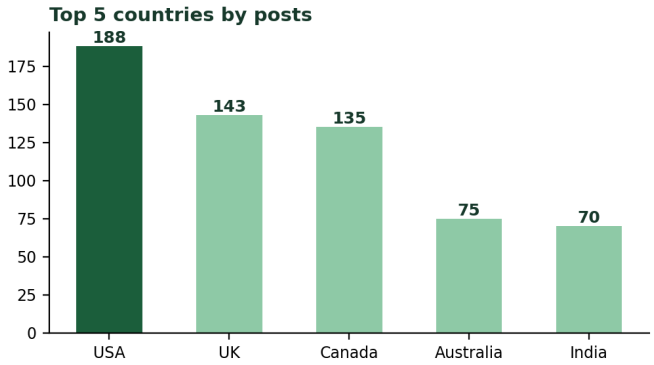




Result: Instagram was both the busiest and the most effective of the three platforms.

3. Geographic Activity

The USA was the most active market with 188 posts and 8,358 likes (about 44.5 likes per post), followed by the UK (143) and Canada (135). Within the USA, Positive had the highest post volume while Neutral earned more likes per post, although the Neutral group is small.



Result: The most active market is not automatically the most engaging, and volume and engagement intensity differ by sentiment.

4. Time-Based Activity & Growth

February (85 posts, 3,505 likes) and 2 PM (94 posts, 3,909 likes) were the busiest month and hour across all years. Both average about 41 likes per post, slightly below the overall 42.9. Post volume surged to 289 in 2023, about 39% of all posts, yet likes per post in 2023 (42.31) stayed close to the overall average.

Result: Audiences are most active at these times, but activity is not the same as engagement, and the 2023 surge in posts did not raise likes per post. Because month and hour results are pooled across years, they are strongly shaped by 2023.

Insights Summary

Dimension	Volume leader	Engagement note
Sentiment	Positive (45 posts)	Joy: 49.3 likes/post vs 37.8
Platform	Instagram (258 posts)	Highest average: 45.1 likes/post
Country	USA (188 posts)	About 44.5 likes/post
Month	February (85 posts)	About 41.2 likes/post
Hour	2 PM (94 posts)	About 41.6 likes/post
Year	2023 (289 posts)	42.31 likes/post vs 42.90 overall

Executive Takeaway

Engagement depends on more than post volume. Joy outperformed Positive, Instagram was the strongest platform, and the USA was the most active market. 2 PM and February were the busiest periods, and the 2023 surge in posts did not raise likes per post.

Recommendations

- **Prioritize Instagram** for high-impact content, as it leads on both volume and likes per post.
- **Build on engaging sentiments**, especially Joy, which earns more engagement per post than Positive.
- **Monitor sentiment in high-activity markets**, particularly the USA.
- **Test posting around 2 PM** and compare likes per post against other hours before committing.

- **Track likes and retweets per post** to measure improvement.

Success Measures

- **Likes per post:** compare engagement quality across platforms, sentiments and years
- **Retweets per post:** monitor how strongly content is shared (current average: 21.51)
- **Sentiment engagement:** check whether Joy keeps outperforming
- **Platform engagement:** compare performance after content changes
- **Peak-period engagement:** test whether 2 PM beats other hours on likes per post

Limitations

- Some sentiment and country-sentiment groups have very few posts, so their averages are indicative only.
- Month and hour results are pooled across years and strongly reflect 2023, which holds about 39% of posts.
- Findings show patterns, not cause and effect.
- Retweets were tracked but not analyzed in depth; this is a next step.

Analyst's Conclusion

Post volume alone does not show performance. Sentiment, platform, geography and timing shape engagement differently, and the 2023 surge in posts did not raise likes per post. These findings support more targeted content planning, stronger engagement strategies and closer sentiment monitoring.